THE MEMORY DUEL: SYNAPTIC PLASTICITY vs. UNBOUNDED KV CACHING

A Comparative Architectural Briefing on Dragon Hatchling (BDH) | DataForge 2026 / NeurIPS Education Track

The Central Falsifiable Claim: A fixed-shape synaptic weight matrix processes sequences of unbounded duration without increasing memory footprint $O(d^2)$, but trades exact token-for-token recall for associative interference blur when fact density exceeds network rank capacity.

1. The Architectural Bottleneck: The KV Cache Memory Wall

Modern autoregressive Transformers retain context by storing every key and value vector in an explicit Key-Value (KV) cache. At sequence length T and hidden dimension d across L layers, memory scales as $O(2 \cdot L \cdot T \cdot d)$. For an 8B parameter model at 128k context, this cache exceeds 16 GB per active stream—creating a severe memory bandwidth bottleneck that dominates inference latency and cost. While eviction, compression, and sliding-window heuristics alleviate immediate RAM limits, they discard intermediate token relationships or induce abrupt context amnesia.

2. The BDH Alternative: Session Memory as Fast Synaptic Plasticity

Pathway's **Dragon Hatchling (BDH)** replaces the growing token buffer with an evolving associative weight matrix $W_t \in$ ■$^{d \times d}$. Instead of comparing a query against all past tokens via quadratic softmax attention, BDH reformulates attention as synaptic plasticity via Hebbian outer-product updates: $\mathbf{W_t = \lambda W_{t-1} + \eta (v_t \otimes k_t}$■$\mathbf{)}$, where $\lambda \in [0, 1]$ governs exponential retention and η is learning rate. Readout occurs via linear associative projection: $\mathbf{y_t = W_t q_t}$. This bounds inference memory strictly at $O(d^2)$—identical whether processing 100 or 10,000,000 tokens—while evaluating per-token inference in strictly constant $O(1)$ FLOPs.

3. Cross-Architectural Comparison

Dimension	Transformer (KV Cache)	Mamba (SSM)	Dragon Hatchling (BDH)
State Footprint	$O(T \cdot d)$ Unbounded Growth	$O(d \cdot N)$ Fixed Latent State	$O(d^2)$ Fixed Synaptic Matrix
Inference Latency	$O(T)$ Memory-Bandwidth Bound	$O(1)$ Recurrent Step	$O(1)$ ReLU-Low-Rank BLAS
Long-Range Recall	Exact token-level matching	Information decay over horizon	Associative Hebbian preservation
Primary Failure Mode	Out-of-Memory (OOM) crash	Loss of associative fidelity	Interference blur under high fact load

4. Scientific Trade-Offs, Failure Modes & Evidence Discipline

The Capacity-Interference Trade-Off: While BDH achieves an infinite theoretical context horizon without allocating new bytes, it obeys the mathematical capacity limits of linear associative memory (Hopfield bounds). If N orthogonal facts are written into a matrix of rank d without decay (λ=1), crosstalk noise scales proportionally to $\sqrt{(N/d)}$. As verified in our interactive testbed, querying Fact #1 after storing 50 dense facts degrades cosine similarity from 0.95 to 0.45 unless decay is tuned. Furthermore, within-session synaptic plasticity must not be confused with permanent parameter weights: fast session memory clears upon conversation reset.

Evidence Status & Industry Integration: Early scaling pretraining experiments for BDH range from 1B to 600B parameters. A specialized GPU-oriented formulation (BDH-GPU) exploits ReLU-low-rank linear attention to avoid recurrent sequential GPU bottlenecks, achieving deployment integration with Amazon SageMaker HyperPod and standard AWS distributed clusters.

5. Primary Scientific Citations (2022–2026)

- *Pathway Research (2025/2026) — The Dragon Hatchling (BDH) Architecture: Biological Synaptic Plasticity as Post-Transformer Session Memory.*
- *Schlag, I., Irie, K., & Schmidhuber, J. — Linear Transformers Are Secretly Fast Weight Programmers (NeurIPS).*
- *Sun, Y., et al. (2023) — Retentive Network: A Successor to Transformer for Large Language Models.*